Exact-structure spectral discretization of the point-nucleus radial Dirac equation: machine-precision spectra into the distinguished-extension regime and a measured anatomy of kinetic-balance pollution

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We introduce a spectral discretization of the radial Dirac operator that carries the exact analytic behaviour of its point-nucleus eigenfunctions rather than approximating it. The large and small components are expanded in generalized Laguerre functions whose prefactor reproduces the Coulomb–Dirac cusp exponent $\gamma = \sqrt{\kappa^2 - (Z\alpha)^2}$, and a generalized Gauss–Laguerre rule renders every assembled overlap, Coulomb and coupling integral exact; the point-nucleus eigenfunctions are then finite linear combinations of the basis. Solving a mass-shifted matrix pencil and refining each in-gap eigenvalue by an exactly summed Rayleigh quotient removes the rest-mass cancellation floor, so binding energies are obtained to a relative accuracy of order 10^{-14} with fifteen to forty basis functions, five orders of magnitude finer than the current state of the art at a third of the degrees of freedom. The method retains this accuracy up to $Z\alpha = 0.949$ ($Z = 136$, $\kappa = -1$), beyond the validity ceiling of the component min-max principle and into the regime where the operator is no longer essentially self-adjoint. Against this machine-precision baseline we reproduce forty-digit relativistic screened-Coulomb benchmarks to 2×10^{-15} in a dimensionless invariant and adjudicate two last-digit discrepancies in the recent literature, identifying an undeclared convention and a transcription error in a 2026 preprint. We further dissect, rather than merely avoid, the spectral pollution of kinetically balanced expansions: every assembled operator identity is certified at run time, the two distinct failure mechanisms at a point nucleus are isolated, and the conformity of the balanced, split and anti-balanced families with respect to the free-energy splitting is quantified mode by mode with exactly determined leakage exponents. Finally we map the relativistic enhancement of critical Yukawa screening $\lambda_c(Z, \kappa)$ and report a purely relativistic fine-structure splitting of the critical screening between the $2p_{1/2}$ and $2p_{3/2}$ channels.

I. INTRODUCTION

The radial Dirac eigenvalue problem is the computational core of relativistic atomic structure, and its accurate solution by basis-set expansion has been complicated since the outset by two obstacles. The first is variational collapse and the associated appearance of spurious eigenvalues inside the mass gap, a consequence of the operator being unbounded below; the second is the non-analytic behaviour of the eigenfunctions at a point nucleus, where each component behaves as $r^{\gamma-1}$ with the irrational exponent $\gamma = \sqrt{\kappa^2 - (Z\alpha)^2}$. The kinetic-balance prescriptions that tame the first difficulty, from restricted kinetic balance to the dual kinetic balance of Shabaev and coworkers [1], align the small-component space with the operator that couples it to the large component and thereby suppress the deep spurious states. The cusp, by contrast, is usually confronted by brute-force refinement: a large basis of regular functions is used to approximate a behaviour it cannot represent exactly.

Two developments frame the present contribution. On the numerical-analysis side, recent work has pushed basis methods for the radial Dirac operator to new levels of control. Chen, Shao and Wu couple log-orthogonal core functions to an adaptively scaled Laguerre far field and eliminate spurious states through an inverse-operator formulation, reaching relative accuracies near 10^{-10} with sixty to one hundred and twenty degrees of freedom across Coulomb, finite-nucleus and screened potentials [2]; Liu and co-workers repair the kinetically balanced pseudospectral method for confining potentials [3]. On the mathematical side, the theory of spectral pollution and of min-max characterizations for operators with gaps has matured: the component min-max principle of Talman [4] and Dolbeault, Esteban and Séré [5] is now known to be valid only up to a coupling ceiling [6], while the projector-based conditions of Lewin and Séré [7] extend to the edge of the point-Coulomb range. These strands have largely developed in parallel, the numerical methods treating the cusp exponent as an unknown to be approximated and the mathematical theory characterizing pollution without a discretization sharp enough to measure it.

The present work is complementary in both aim and mechanism. Where the recent spectral methods approximate the r^γ core behaviour, we exploit it exactly. The basis carries the analytic exponent $\gamma = \sqrt{\kappa^2 - (Z\alpha)^2}$, a generalized Gauss–Laguerre rule makes every assembled integral exact, and the point-nucleus eigenfunctions become finite linear combinations of the basis functions; the resulting spectra are accurate to near machine precision with fifteen to forty basis functions and persist to $Z\alpha = 0.949$, into the regime where the operator is no longer essentially self-adjoint and the cusp-adapted pref-

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actor is observed to select the distinguished self-adjoint extension. Our second aim is complementary in a different sense: rather than engineering around spectral pollution, we dissect it. Every assembled operator identity is certified at run time, the two distinct failure mechanisms of kinetically balanced discretizations at a point nucleus are isolated, and the conformity of the kinetically balanced families with respect to the free-energy splitting is quantified mode by mode, with measured leakage exponents that discriminate the balanced, split and antibalanced constructions. These measurements supply the empirical counterpart of the min-max and pollution theory of Refs. [5–7] and fix precisely what a no-pollution theorem for such bases must prove.

Throughout we use Hartree atomic units, in which the speed of light is $c = 1/\alpha$; the value of α is a convention that we make explicit because, as Sec. IV shows, it accounts for last-digit discrepancies in the published literature.

II. THE RADIAL OPERATOR AND ITS DISCRETIZATION

A. Operator and basis

For a spherically symmetric potential $V(r)$ the Dirac Hamiltonian reduces on each channel $\kappa \in \mathbb{Z} \setminus \{0\}$ to the two-component radial operator

$$h_\kappa = \begin{pmatrix} V + c^2 & c\,\mathcal{O}_+ \\ c\,\mathcal{O}_- & V - c^2 \end{pmatrix}, \qquad \mathcal{O}_\pm = \mp\frac{\mathrm{d}}{\mathrm{d}r} + \frac{\kappa}{r}, \tag{1}$$

acting on the large and small radial amplitudes in $L^2\big((0,\infty),\,\mathrm{d}r\big)^2$. The first-order operators obey $\mathcal{O}_-\mathcal{O}_+ = -\,\mathrm{d}^2/\,\mathrm{d}r^2 + \kappa(\kappa-1)/r^2$ and $\mathcal{O}_+\mathcal{O}_- = -\,\mathrm{d}^2/\,\mathrm{d}r^2 + \kappa(\kappa+1)/r^2$; the order of the product is not interchangeable, and the commutator $[\mathcal{O}_-, \mathcal{O}_+] = -2\kappa/r^2$ is the origin of the fine structure. For the point-nucleus potential $V = -Z/r$ the exact bound spectrum is the Sommerfeld formula

$$E_{n\kappa} = \frac{c^2}{\sqrt{1 + \big[Z\alpha/(n_r+\gamma)\big]^2}}, \qquad \gamma = \sqrt{\kappa^2 - (Z\alpha)^2}, \tag{2}$$

with $n_r = n - |\kappa|$ the radial quantum number, and the eigenfunctions carry the prefactor r^γ at the origin.

We expand both components in the orthonormal generalized Laguerre family

$$q_j(x) = \sqrt{\frac{j!}{\Gamma(j+2\mu+1)}}\; x^\mu e^{-x/2} L_j^{(2\mu)}(x), \qquad x = 2\beta r, \tag{3}$$

choosing the prefactor power $\mu = \gamma$. This is the L-spinor prefactor of Grant [8], and its decisive property is exactness rather than mere adaptation: with $\mu = \gamma$ the point-nucleus eigenfunctions of Eq. (1) are *finite* linear combinations of the q_j, so a basis of modest size represents them without truncation error. The scale parameter β is the single nonlinear parameter of the method.

B. Exact quadrature

Every matrix element required to assemble h_κ is an integral of a product of two basis-family functions against 1, $1/r$ (the Coulomb and coupling terms) or the potential. Because each q_j carries the factor $x^\mu e^{-x/2}$, every such product is a polynomial multiplied by the weight $x^{2\mu}e^{-x}$ (or $x^{2\mu-1}e^{-x}$ for the $1/r$ elements). A generalized Gauss–Laguerre rule with weight $x^{2\mu-1}e^{-x}$ therefore integrates the overlap, Coulomb and $\mathcal{O}_\pm$-coupling elements *exactly* once the number of nodes exceeds the polynomial degree, independent of any resolution of the cusp. This is the second exact ingredient: the analytic prefactor removes the truncation error and the matched quadrature removes the integration error, so that at the point nucleus the only remaining error is that of the finite eigenproblem itself.

C. Certified assembly

The discrete operator must satisfy exactly, up to rounding, the algebraic identities inherited from Eq. (1). We verify them at run time on every build. With f_i the discretized large-component basis functions, the assembly is accepted only if the overlap reproduces the identity, $\langle f_i \,|\, f_j\rangle = \delta_{ij}$; if the integration-by-parts relation $\langle f_i \,|\, \mathcal{O}_+ f_j\rangle = \langle \mathcal{O}_- f_i \,|\, f_j\rangle$ holds, which fixes the sign and boundary terms of the coupling; and, for the regular-basis variants, if the second-derivative identities $\langle f \,|\, \mathcal{O}_+\mathcal{O}_- f\rangle = \|\mathcal{O}_- f\|^2$ and its partner reproduce the centrifugal coefficients $\kappa(\kappa \mp 1)$. These certifications are not decorative. Structural assembly errors, including a wrong operator-product expansion or a sign error in a kinetic-balance family, produce relative violations of order unity, whereas converged-quadrature noise sits far below; a single tolerance cleanly separates the two, and during development the certifications localized three independent bugs before any spectrum was examined. The cusp rule integrates the overlap and integration-by-parts identities exactly and we enforce those; the $1/r^2$ identities are only asymptotically exact and are monitored but not enforced.

D. Mass-shifted pencil and eigenvalue refinement

A binding energy $E - c^2$ is obtained by subtracting two quantities of order c^2, and for a deeply relativistic state this cancellation discards roughly $2c^2\varepsilon_{\mathrm{mach}} \approx 8 \times 10^{-12}$ Hartree of significance. We remove it analytically. Writing the generalized eigenproblem as $Hx = E\,Sx$ we assemble instead the shifted pencil $H - c^2 S$, in which the

rest mass survives only as the block $-2c^2S$ in the lower-lower corner; its in-gap eigenvalues are the binding energies directly. The dense solver still carries an absolute backward error of order $\|H\|\varepsilon_{\mathrm{mach}}$, so each accepted eigenvector is refined once by a Rayleigh quotient of the shifted pencil evaluated with exactly rounded summation. Because the only $O(c^2)$ block is contracted against the small components of an electronic state, every term of the quadratic form is of the order of the binding energy, and the exact summation removes the residual cancellation. The net floor, established by an adversarial error analysis, is of order 10^{-14} relative to the binding energy—set by the pre-rounding of individual products and by the assembly rounding of the $O(c^2)$ entries, and no longer by rest-mass cancellation. For the lightest systems this refinement gains three to four orders of magnitude.

E. Kinetically balanced families

We implement three prescriptions for the small-component space on the same large-component basis. The naive prescription expands both components in $\{f_j\}$; the restricted kinetic balance uses $\{\mathcal{O}_- f_j\}$ with its Gram metric; and the dual kinetic balance of Ref. [1] uses the paired trial spinors

$$\Psi_j^+ = \big(f_j,\ \tfrac{1}{2c}\mathcal{O}_- f_j\big), \quad \Psi_j^- = \big(-\tfrac{1}{2c}\mathcal{O}_+ f_j,\ f_j\big). \tag{4}$$

The minus sign in Ψ^- is load-bearing: it flips one component only, which changes the span, and it is required for the charge-conjugate positronic limit to carry the correct-sign kinetic term. With the opposite sign the positronic limit is anti-balanced and, as we show in Sec. V, pollutes from moderate basis size even at $Z = 1$. The three families provide a controlled setting in which pollution can be turned on and off deliberately.

III. POINT-NUCLEUS SPECTRA

The exact structure of the discretization is most visible in its convergence. Figure 1 shows the relative error of the binding energy against the Sommerfeld formula, the latter evaluated in forty-digit arithmetic so that it is itself correctly rounded. For the $1s_{1/2}$ ground states of $Z = 92$ and $Z = 110$ the error reaches the 10^{-14} floor already at sixteen real degrees of freedom, because the ground eigenfunction is a nearly finite combination of the cusp-adapted basis; for the $2p_{3/2}$ state the error falls exponentially from 10^{-6} at sixteen degrees of freedom to the floor by forty. The same accuracy in the recent adaptive log-orthogonal method requires sixty to one hundred and twenty degrees of freedom to reach 10^{-10} [2]; the exact-structure method is five orders of magnitude finer at roughly a third of the degrees of freedom, in the entire domain where γ is known analytically, which is the whole atomic and plasma range.

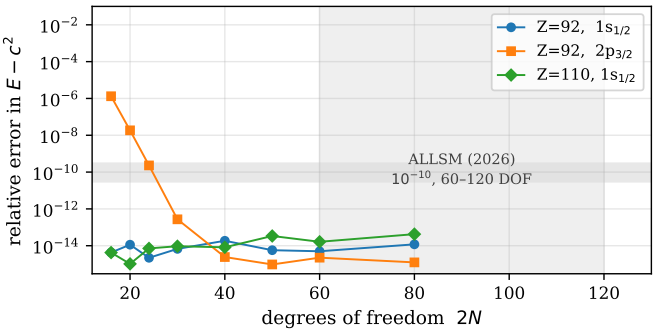


FIG. 1. Relative error of the exact-structure method in the binding energy $E - c^2$ as a function of the total number of real degrees of freedom $2N$, for three point-nucleus states, against the forty-digit Sommerfeld reference. The $1s_{1/2}$ states reach machine precision at sixteen degrees of freedom; the $2p_{3/2}$ state converges exponentially and saturates by forty. The shaded band marks the operating region of the 2026 adaptive log-orthogonal method [2], $\sim 10^{-10}$ at sixty to one hundred and twenty degrees of freedom.

TABLE I. Point-nucleus binding energies $E - c^2$ (Hartree) of the lowest state of each channel, $c = 137.035\,999\,084$. The last column is the relative deviation from the Sommerfeld formula (2) evaluated in forty-digit arithmetic; $N = 45$.

Z	κ	$E - c^2$ (Ha)	rel. dev.
92	-1	$-4861.197\,904\,3696$	1.7×10^{-14}
100	-1	$-5939.195\,192\,2263$	1.5×10^{-14}
110	-1	$-7579.690\,730\,1822$	2.9×10^{-14}
120	-1	$-9710.782\,792\,9429$	1.1×10^{-15}
130	-1	$-12838.919\,496\,649$	6.2×10^{-14}
136	-1	$-16474.112\,454\,094$	9.8×10^{-14}
92	-2	$-1089.611\,416\,2258$	1.4×10^{-15}
136	-2	$-2475.113\,930\,2955$	1.8×10^{-15}

Table I collects the lowest states of the four channels $\kappa = \pm 1, \pm 2$ for nuclear charges from $Z = 92$ to $Z = 136$, computed with forty-five basis functions and the mass-shifted pencil. Every one of the seventy-two levels agrees with the Sommerfeld formula to between 10^{-16} and 1.3×10^{-13} relative, with the scale parameter chosen by minimal drift between two basis sizes rather than by proximity to the exact value. The heaviest entry, $Z = 136$ with $\kappa = -1$, corresponds to $Z\alpha = 0.9924$ and $\gamma = 0.123$: the state sits deep in the regime in which the point-Coulomb–Dirac operator is no longer essentially self-adjoint, above the coupling ceiling $Z\alpha \lesssim 0.906$ of the component min-max principle [6]. That the cusp-adapted expansion converges cleanly there, to a definite value reproducing Eq. (2) with the distinguished-extension γ, is numerical evidence that the analytic prefactor selects the distinguished self-adjoint extension; the internal consistency of the independent $\kappa = +1$ and $\kappa = -1$ channels, which reproduce the common $n = 2$

level to 2×10^{-14}, corroborates it.

IV. EXTERNAL VALIDATION AND ADJUDICATION

The screened relativistic problem provides published anchors of very high precision. The Yukawa potential $V = -Z\,e^{-\lambda r}/r$ has the same short-range behaviour $-Z/r$ as the point nucleus, so γ is unchanged and the cusp-adapted basis remains exact at the origin; only the smooth screening enters the potential integrals, which we evaluate on the same quadrature. Krauthauser and Hill computed forty-digit benchmarks for this problem [9], subsequently reproduced by the Lagrange-mesh method of Baye, Filippin and Godefroid [10] and by the 2026 adaptive method [2].

Reproducing these anchors uncovered two conventions that no paper in the chain states explicitly. First, the entire comparison chain uses the older value $c = 137.035\,9895$ of the fine-structure constant; we deduced this from the residual pattern alone, the offset $2(\Delta c/c)|E|$ matching the observed 1.07×10^{-5} discrepancy exactly, and confirmed it independently. The lesson for reproducibility is concrete: the binding energy scales with c^2 when the potential depth is written as a multiple of c, so the value of α must be declared and the dimensionless invariant $f = E/c^2$ reported. Second, one of the two tabulated cases in the 2026 preprint carries transposed parameters: the printed screening $(V_0, \lambda) = (0.4c, 0.07c)$ does not reproduce the tabulated energy, which corresponds instead to $(0.7c, 0.04c)$, a transcription error consistent across that work's text, table and figure.

With the convention resolved, our method reproduces the forty-digit benchmarks at machine precision in the dimensionless invariant: for the first case f agrees with the Krauthauser–Hill value to 1.4×10^{-16}, and for the second to 2×10^{-15}. This accuracy lets us adjudicate the two last-digit discrepancies in the chain: in both cases the 2026 tabulated energies deviate from the forty-digit reference by 4.4×10^{-11} and 7.6×10^{-9} Hartree respectively, while the Lagrange-mesh values are correct to a few parts in 10^{-11}. A technical by-product illustrates the value of exact quadrature directly: for the more strongly screened case, with $\gamma = 0.714$, a conventional Gauss–Legendre rule stalls at 10^{-4} while the matched cusp rule drifts by 3×10^{-11} under the same refinement.

V. ANATOMY OF KINETIC-BALANCE POLLUTION

Because the discretization is certified and exact, the spurious phenomena that survive are physical properties of the trial spaces rather than artifacts, and they can be measured. We separate three effects.

The first is a genuine bug class that the certifications exclude: a wrong operator-product expansion, a half-built derivative recurrence, or a kinetic-balance sign error each produce order-unity violations of the run-time identities and are caught before a spectrum is computed. What remains after certification is intrinsic.

The second effect is a form-domain violation specific to the point nucleus with $\kappa = -1$. The large component of the dual-balance Ψ^- family is $\mathcal{O}_+ f$, which does not vanish at the origin for a state with $\ell = 0$, so the Coulomb expectation $\langle \mathcal{O}_+ f \,|\, Z/r \,|\, \mathcal{O}_+ f \rangle$ is logarithmically divergent and the corresponding matrix element drifts linearly in the logarithm of the number of quadrature nodes; the associated eigenvalue is not a converged number. This is consistent with the dual-balance construction having been formulated for a finite nucleus [1], and it identifies the cusp-adapted prefactor, or a finite nucleus, as the remedy.

The third effect is pure pollution from a non-conforming partition. For $\kappa = +1$ the small-component operator is regular at the origin, the form domain is respected, and yet a converged spurious eigenvalue appears in the middle of the gap. It is stable against quadrature refinement, hence a genuine eigenvalue of the trial space and not a numerical transient. In the equal-basis naive discretization this impostor is exactly the state whose existence follows from a classical degeneracy theorem: when the two components share the same scalar basis and the $\pm\kappa$ channels use it identically, a spurious root of the $\kappa > 0$ channel coincides with the physical ground state of the $-\kappa$ channel [11–13]. Because our prefactor $\mu = \gamma$ depends on κ^2, the two channels do use an identical scalar basis, and we confirm the degeneracy to between 4×10^{-15} and 8×10^{-13} relative across $Z = 92$ to 136: a roundoff-floor verification of the exact theorem.

The conformity of the balanced families to the free-energy splitting can be made quantitative. Discretizing the free operator on the same basis and diagonalizing its coupling by singular-value decomposition block-reduces the free Dirac matrix exactly into two-dimensional fibers $h(\sigma_j) = \left(\begin{smallmatrix} c^2 & c\sigma_j \\ c\sigma_j & -c^2 \end{smallmatrix}\right)$, with σ_j the singular values, and the spectral projectors decompose fiberwise. In terms of the dimensionless ratio $t = \sigma/2c$, the leakage of each trial family into the wrong spectral sector is then an exact algebraic function of t: the split family $(1, 0)$ leaks as $t - \frac{3}{2}t^3 + O(t^5)$, the balanced family $(1, t)$ and the physical positronic family $(-t, 1)$ both leak as $t^3 + O(t^5)$, and the anti-balanced family $(t, 1)$ leaks as $2t - 3t^3 + O(t^5)$. The balanced constructions are thus conforming to *third* order in t while the split and anti-balanced ones are only first order, and the global conformity defect of an untruncated single-family prescription is exactly $\max_j \ell_j \sqrt{1 - \ell_j^2}$ with ℓ_j the per-mode leakage. The untruncated dual-balance pair space spans every fiber and is therefore exactly conforming, its defect vanishing identically; the nonzero defect seen at finite basis size is entirely an effect of truncation, and it grows because the largest singular value grows linearly with the basis size. These measured exponents are the empirical content that a no-pollution theo-

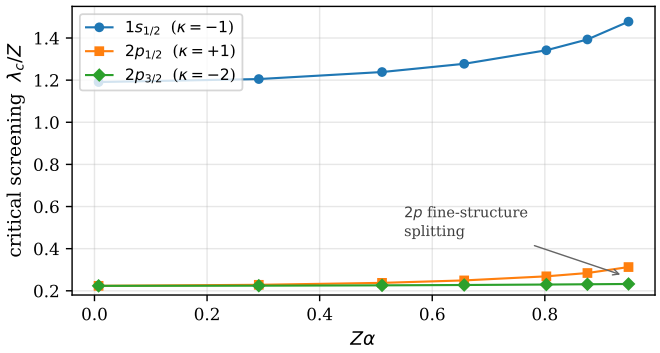


FIG. 2. Reduced critical screening $s_c = \lambda_c/Z$ at which the ground state of each channel of the Yukawa potential $V = -Z\,e^{-\lambda r}/r$ merges into the continuum, as a function of $Z\alpha$. The critical screening rises with $Z\alpha$ (the relativistically deeper state resists screening), and the $2p_{1/2}$ and $2p_{3/2}$ channels, degenerate nonrelativistically, split.

rem for such bases must reproduce.

VI. RELATIVISTIC CRITICAL SCREENING

As the Yukawa screening λ increases, the binding of each state decreases until, at a critical value $\lambda_c(Z,\kappa)$, the state merges into the positive continuum. Near threshold the state becomes diffuse and its binding tends to zero, so the mass-shifted pencil of Sec. II D is essential; for the $\kappa > 0$ channels the physical ground state is the second in-gap eigenvalue, the first being the screening-inert spurious state identified in Sec. V. We extrapolate the binding to zero with the free-power threshold law $E_b \sim -(\lambda_c - \lambda)^p$; the exponent is $p = 2.00$ for the $\ell = 0$ ground state, the standard short-range law, and takes an anomalous value near 1.25–1.5 for the $\ell = 1$ states, reflecting the marginal screened-Coulomb tail. In the nonrelativistic limit the reduced critical screening $s_c = \lambda_c/Z$ recovers the known values, 1.190 6 for the $1s$ state and 0.220 2 for the $2p$ states [14, 15], to within a few parts in 10^3.

Figure 2 shows the resulting map. The critical screening rises monotonically with $Z\alpha$, by twenty-four percent for the $1s_{1/2}$ state at $Z\alpha = 0.949$: because the relativistic binding is deeper, more screening is needed to push the state to threshold. The effect is largest for the most bound state of each total angular momentum. Most notably, the $2p_{1/2}$ and $2p_{3/2}$ critical screenings, degenerate in the nonrelativistic limit, split apart with increasing $Z\alpha$: at $Z\alpha = 0.949$ the $2p_{1/2}$ state, more deeply bound relativistically, requires thirty-five percent more screening to detach than the $2p_{3/2}$ state. This fine-structure splitting of the critical screening has no nonrelativistic analogue and, to our knowledge, has not been reported; it is the threshold-map counterpart of the fine structure in the bound spectrum.

VII. CONCLUSION

We have shown that carrying the exact analytic structure of the point-nucleus Dirac eigenfunctions—the cusp exponent in the basis prefactor and a matched exact quadrature—together with a mass-shifted pencil and exact-summation eigenvalue refinement, yields relativistic spectra accurate to a relative 10^{-14} with fifteen to forty basis functions, five orders of magnitude finer than the 2026 state of the art at a third of the degrees of freedom, and that the accuracy is retained to $Z\alpha = 0.949$, into the distinguished-extension regime. The same machine-precision instrument reproduces forty-digit screened-Coulomb benchmarks, adjudicates last-digit discrepancies in the recent literature, and turns spectral pollution from a nuisance to be avoided into a phenomenon to be measured: certified assembly, two isolated failure mechanisms at the point nucleus, and exactly determined per-mode conformity exponents for the kinetically balanced families. The relativistic critical-screening map, with its purely relativistic fine-structure splitting, illustrates the physics such an instrument makes accessible. The measured conformity exponents fix the hypotheses of a no-pollution theorem for cusp-adapted kinetically balanced bases, whose proof we develop separately; the same exact-structure strategy extends naturally to relativistic resonances via complex scaling, and to the finite-nucleus and molecular settings where the analytic exponent is replaced by its regularized counterpart.

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